

Human-AI Interactions in Craftwork Environments

Samantha Pan, Benjamin Liang, Oliver Lee, Hannah Li, Sana Purdhani, Rui Yang, Julia Zurek

Mentors: Christine Marcelino, Kyungwon Koh, PhD, Alice Shinhye Cha, and Elena Zilar

School of Information Sciences, Grainger College of Engineering, College of Fine & Applied Arts, College of Liberal Arts & Sciences, University of Illinois Urbana-Champaign

Introduction

How can we intentionally cultivate human capacities through interaction with artificial intelligence technologies? We aim to use a hands-on craft workshop to answer our research questions through observing how humans interact with AI tools. Our research emphasizes human creativity, AI influence, emotional intelligence, and tacit knowledge.

Research Goals

- Identify which human capacities remain central while working alongside AI tools.
- Assess how AI tools influence **decision making, creativity**, and **confidence** in the making process.
- Analyze how **emotions** can shape and influence the making process.
- Examine the role of **tacit knowledge** and **embodied intelligence** in the making process.

Questions

- How do participants **co-create** with AI during the making process?
- What forms of **tacit knowledge** or **embodied intelligence** emerge through this co-making process?
- What tensions, insights, or shifts in practice arise when making with AI?
- What **emotions** are apparent in each step of the making process?

Key Concepts

Constructionism

- Learning through making tangible artifacts
- This approach is useful for creating and critically examining AI systems (Kahn & Winters, 2021)

Experiential learning

- Learning by doing
- Emphasizes unique human abilities necessary for us to ethically and creatively engage with AI tools (Jayaraman, 2023)

Tacit knowledge

- Embodied, experiential knowledge that emerges through working with materials and processes

Ontological design

- Tools and technologies we create shape how we think, act, and make, and we reshape them through use

Embodied Intelligence

- Intelligence built through information perception and interaction with the environment over time (Liu et al, 2025)

Design Based Research

Design based research (**DBR**) is an **iterative** and **mixed-methods** process of **designing solutions** to real-world problems, tested in **real-world settings**.

Iterative: constant testing and refining of research methods and conclusions, using data to continuously improve decisions.

Our research project is a pilot study, the first iteration. We will use e-textile crafts and AI to gather preliminary data that will inform future in-depth research on our questions.

Research Design

Study Setting:

2-hour e-textile patch workshop at the CU Community Fab Lab (April 2026)

Participants:

Purposive sample of 10-15 volunteers of diverse AI and making skill levels

Data Collection:

- Pre and Post Surveys
- Video/audio
- Chatbot history
- Facilitator observations
 - Points of confusion, realization, and questions
- Post workshop Interviews

Data Analysis:

- Skill level vs. Results
- Chatbot Response vs. Design Result
- Qualitative data
 - Ease of creation + understanding
 - Effectiveness of AI responses
 - Overlap of emotional responses between participants

Preliminary Insights

- AI might increase engagement through collaborative brainstorming
- AI is expected to act as a supportive collaborator and not a replacement
- In maker environments, people tend to turn to peers rather than AI
- Easier to learn from visual demonstration
- Skill level vs. benefits of AI use



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